

Exam: Sheet 1
Level:
Course Title: Intelligent Information Systems



Course Code:
Date:
Time:

Q1 Demonstrate information value concept.

Answer:

The value of information is directly linked to how it helps decision makers achieve their organization's goals. Valuable information can help people perform tasks more efficiently and effectively. Many businesses assume that reports are based on correct, quality information, but, unfortunately, that is not always true.

Q2 Explain Accessible, Accurate, Complete, and Flexible Characteristic.

Answer:

Accessible: Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.

Accurate: Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. This is commonly called garbage in, garbage out.

Complete: Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.

Flexible: Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the amount of money the company has invested in inventory.

Q3 Explain Relevant, Reliable, Timely, and Verifiable Characteristic.

Answer:

Relevant: Relevant information is important to the decision maker. Information showing that lumber prices might drop is probably not relevant to a computer chip manufacturer.

Reliable: Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up may not be reliable.

Timely: Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.

Verifiable: Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

Q4 Define an information system (IS).

Answer:

An **information system (IS)** is a set of interrelated components that collect, process, store, and disseminate data and information; an information system provides a feedback mechanism to monitor and control its operation to make sure it continues to meet its goals and objectives.

Q5 Demonstrate three of CBIS components.

Answer:

People make the difference between success and failure in all organizations. **Those who build great companies understand that the ultimate throttle on growth for any great company is not markets, or technology, or competition, or products. It is one thing above all others: the ability to get and keep enough of the right people.** Thus, it comes as no surprise that people are the most important element in computer-based information systems.

Good systems can enable people to produce exceptional results. They can also boost job satisfaction and worker productivity. Information systems personnel include all the people who manage, run, program, and maintain the system, including the chief information officer (CIO), who leads the IS organization. End users are people who work directly with information systems to get results. They include financial executives, marketing representatives, and manufacturing line operators.

A **procedure** defines the steps to follow to achieve a specific end result, such as enter a customer order, pay a supplier invoice, or request a current inventory report. Good procedures describe how to achieve the desired end result, who does what and when, and what to do in the event something goes wrong. When people are well trained and follow effective procedures, they can get work done faster, cut costs, make better use of resources, and more easily adapt to change.

Q6 explain each of Well-trained workers, System support, Better teamwork, and Redesigned processes.

Answer:

- **Well-trained workers.** Employees must be well trained and understand the need for the new system, what their role is in using or operating the system, and how to get the results they need from the system.
- **System support.** Trained and experienced users who can show others how to gain value from the system and overcome start-up problems.
- **Better teamwork.** Employees must understand and be motivated to work together to achieve the anticipated benefits of the system.
- **Redesigned processes.** New systems often require radical redesign of existing work processes as well as the automation of new processes.

Q7 Demonstrate of Units of measure for data.

Answer:

Unit of Measure	Size
Byte	1 byte
Kilobyte	1,000 bytes
Megabyte	1,000 kilobytes
Gigabyte	1,000 megabytes
Terabyte	1,000 gigabytes
Petabyte	1,000 terabytes
Exabyte	1,000 petabytes
Zettabyte	1,000 exabytes
Yottabyte	1,000 zettabytes

Q8 Demonstrate each of Internet of Things (IoT) and Internet of Everything (IoE).

Answer:

The **Internet of Things (IoT)** is a network of physical objects or “things” embedded with sensors, processors, software, and network connectivity capability to enable them to exchange data with the manufacturer of the device, device operators, and other connected devices. In theory, the IoT would enable us to connect almost any device with an on/off switch to a network—automobiles, appliances, components of an aircraft engine, heart monitor implants, packing labels, ingestible pills, wearable devices, and even highway sensors that can warn of traffic and hazardous road conditions.

The **Internet of Everything (IoE)** encompasses not only machine-to-machine but also people-to-people and people-to-machine connections. It is estimated that the total number of devices supported by the IoE could reach 50 billion by the end of 2020. This rapid growth is being fueled by the increasing availability of network access, the creation of more inexpensive smart devices with sensors and network capabilities built into them, the rapid growth in smartphone penetration, and the creativity and innovation of people who are able to see and capitalize on the almost unlimited opportunities.

With my best wishes

Prof. Dr. Mohamed Okasha